

Lab 3

Write a query to find out which subjects are not associated with any track.

```
postgres@lol:~$ psql
psql (14.17 (Ubuntu 14.17-0ubuntu0.22.04.1))
Type "help" for help.

postgres=# \c advpostgres
You are now connected to database "advpostgres" as user "postgres".
advpostgres=# SELECT sub_name
FROM subject
WHERE id NOT IN (
    SELECT sub_id FROM track_sub
);
 sub_name
-----
(0 rows)

advpostgres=#
```

Display name and age of each students

```
advpostgres=# SELECT e_name, EXTRACT(YEAR FROM AGE(birth_date)) AS age
FROM student;
 e_name | age
-----+-----
Ahmed Youssef | 26
Sara Mahmoud | 24
Mohamed Ali | 26
Nourhan Adel | 24
Khaled Saeed | 27
(5 rows)

advpostgres=#
```

Display the name of students with their rounded score in each subject

```
(5 rows)
advpostgres=# SELECT s.e_name, sub.sub_name, ROUND(g.grade) AS rounded_score
FROM grades g
JOIN student s ON g.stu_id = s.id
JOIN subject sub ON g.sub_id = sub.id;
 e_name      | sub_name      | rounded_score
-----+-----+-----
Ahmed Youssef | Databases     |           96
Sara Mahmoud  | Networks      |           88
Mohamed Ali   | Programming    |           77
Nourhan Adel  | Machine Learning |           91
Khaled Saeed  | Operating Systems |           82
(5 rows)

advpostgres=#
```

Display the name of students with the year of Birthdate

```
advpostgres=# SELECT e_name, EXTRACT(YEAR FROM birth_date) AS birth_year
FROM student;
 e_name      | birth_year
-----+-----
Ahmed Youssef |      1999
Sara Mahmoud  |      2000
Mohamed Ali   |      1998
Nourhan Adel  |      2001
Khaled Saeed  |      1997
(5 rows)

advpostgres=#
```

Add new exam result, in date column use NOW() function

```
advpostgres=# INSERT INTO exam (date) VALUES (NOW());
INSERT 0 1
advpostgres=#
```

Write a query to calculate the average grade obtained by a specific student across all exams.

```
advpostgres=# SELECT AVG(grade) AS avg_grade
FROM grades
WHERE stu_id = 1;
 avg_grade
-----
95.5000000000000000
(1 row)
```

Write a query to replace all occurrences of 'gmail.com' in email addresses

with 'iti.com'.

```
advpostgres=# UPDATE student
SET email = REPLACE(email, 'gmail.com', 'iti.com')
WHERE email LIKE '%@gmail.com';
UPDATE 5
advpostgres=#
```

```
advpostgres=# SELECT e_name, email
FROM student
WHERE email LIKE '%@iti.com';
 e_name      | email
-----+-----
Ahmed Youssef | ahmed.youssef@iti.com
Sara Mahmoud  | sara.mahmoud@iti.com
Mohamed Ali   | mohamed.ali@iti.com
Nourhan Adel  | nourhan.adel@iti.com
Khaled Saeed  | khaled.saeed@iti.com
(5 rows)
advpostgres=#
```

Write a query to calculate the difference in days between the current date and each exam date.

```
advpostgres=# SELECT id AS exam_id, date AS exam_date,
CURRENT_DATE - date AS days_difference
FROM exam;
 exam_id | exam_date | days_difference
-----+-----+-----
1 | 2024-05-01 | 364
2 | 2024-05-05 | 360
3 | 2024-05-10 | 355
4 | 2024-05-15 | 350
5 | 2024-05-20 | 345
6 | 2025-04-30 | 0
(6 rows)
```

Write a query to check if each student's email address ends with '.com'.

```
advpostgres=# SELECT e_name, email,
CASE
    WHEN email LIKE '%.com' THEN 'Yes'
    ELSE 'No'
END AS ends_with_com
FROM student;
 e_name      | email              | ends_with_com
-----+-----+-----
Ahmed Youssef | ahmed.youssef@iti.com | Yes
Sara Mahmoud  | sara.mahmoud@iti.com | Yes
Mohamed Ali   | mohamed.ali@iti.com  | Yes
Nourhan Adel  | nourhan.adel@iti.com | Yes
Khaled Saeed  | khaled.saeed@iti.com | Yes
(5 rows)
```

Display each exam date like 'MM/DD/YYYY'.

```
advpostgres=# SELECT id AS exam_id,  
      TO_CHAR(date, 'MM/DD/YYYY') AS formatted_date  
FROM exam;  
 exam_id | formatted_date  
-----+-----  
      1 | 05/01/2024  
      2 | 05/05/2024  
      3 | 05/10/2024  
      4 | 05/15/2024  
      5 | 05/20/2024  
      6 | 04/30/2025  
(6 rows)
```